SPE-229571-MS

Agentic AI on Customer Dashboards in Remote Monitoring and Diagnostic Services

Laura Nuti and Giacomo Veneri, IET GTS Engineering Digital and iCenter™ - Baker Hughes, Florence, Italy; Daniele Porciani, IET GTS Digital - Baker Hughes, Florence, Italy; Leonardo Catalano, Domenico Iasparro, Michele Lauriola, and Marco Capanni, IET GTS Engineering Digital and iCenter™ - Baker Hughes, Florence, Italy; Mauro Vitale, IET AI Team - Baker Hughes, Florence, Italy

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

The Oil & Gas industry is increasingly confronted with stringent requirements related to process reliability, operational efficiency, and maintenance cost reduction. In this context, companies are actively seeking advanced technological solutions capable of monitoring the performance of critical process equipment and enabling rapid, informed decision-making.

Agentic AI systems offer a promising response to these demands by embedding autonomous, goal-directed intelligence into digital service platforms. These systems not only interpret and synthesize data from multiple sources but also provide contextualized insights and proactive recommendations through natural language interaction. Their ability to support fast and accurate diagnostics, reduce time spent on manual analysis, and minimize equipment downtime positions them as strategic enablers in the pursuit of operational excellence and cost-effective maintenance.

Remote Monitoring & Diagnostics (RM&D) systems for high-performance turbomachinery traditionally rely on passive Customer Dashboards that offer a wealth of valuable information that can significantly aid in analyzing the operational conditions and health of equipment. However, retrieving and interpreting this data is often challenging due to the presence of multiple tabs and the complexity of the analyses required to extract meaningful insights. Conducting accurate analyses may demand expert knowledge, and the time spent navigating the dashboard or identifying the appropriate analytical approach could be better allocated to more strategic tasks. Many users may experience a sense of confusion or frustration when unable to locate the information they need. To address these challenges, we propose the integration of single-agentic AI systems capable of aggregating, summarizing, and connecting data from multiple sources, thereby streamlining the analytical process and significantly reducing cognitive load and accelerating decision-making in critical turbomachinery applications. The proposed system addresses critical security challenges through a novel architectural approach that completely decouples authorization logic from AI components via deterministic orchestration layers. This implementation demonstrates that organizations can deploy advanced AI capabilities in sensitive industrial environments without compromising intellectual property

or data confidentiality. Our system transforms operator interaction from static data consumption to dynamic dialogue and establishes a foundation for secure AI integration in industrial monitoring systems while providing practical guidance for enterprise AI deployment in regulated environments.

Introduction

High-performance turbomachinery is fundamental to the operation of key sectors including energy, fertilizer production, and refining. System failures in these applications can lead to substantial production losses, increased emissions, and significant financial impacts. The criticality of these assets necessitates robust maintenance, monitoring and diagnostic strategies to prevent failures and identify and resolve issue before they arise [1].

Remote Monitoring & Diagnostics (RM&D) services are thus central to managing the performance, reliability, and maintenance of complex industrial systems. However, the value of these systems often hinges on how efficiently users can interpret and act on vast streams of data. Traditionally, customer dashboards have served as passive visualization tools, presenting raw sensor data, trend lines, and alerts. These dashboards demand technical expertise and time-intensive manual analysis to translate data into meaningful decisions.

Agentic AI systems [15] introduce a fundamental shift in the RM&D landscape by embedding autonomous, goal-directed intelligence directly into dashboard interfaces. Rather than simply displaying telemetry data, these systems actively monitor, interpret, and summarize operational conditions. They function as virtual analysts—capable of detecting anomalies, forecasting potential failures, and communicating insights through natural, conversational language.

Traditional interactions with RM&D data typically require trained users to manually retrieve, interpret, and analyze information. In contrast, agentic AI enables users to access all relevant data—whether available through web portals or distributed via email—in an integrated and intuitive manner, using natural language interaction. This transition transforms the experience from static data consumption to dynamic dialogue with the system.

By converting complex datasets into language-based insights, agentic AI systems elevate RM&D dashboards from passive monitoring tools to intelligent collaborators in operational decision-making. This evolution reduces cognitive load, accelerates decision-making, and enhances both accessibility and scalability of support services. The paper presents a proof-of-concept implementation of an AI Assistant for Digital Services, illustrating the potential of agentic AI to redefine industrial diagnostics and performance optimization through human-AI collaboration. This evolution supports a future where human-AI collaboration becomes standard in industrial diagnostics and performance optimization [2].

The presented paper explores the expansion of capabilities for RM&D Digital Services enabled by Agentic AI systems. Among the key advancements is the enhancement of Natural Language Understanding (NLU), which allows the AI assistant to process complex queries, manage follow-up questions, and interpret contextual references. This includes support for multilingual interaction and the comprehension of domain-specific technical terminology, significantly improving user accessibility and interaction quality.

Another notable capability is the system's ability to deliver proactive recommendations. Rather than responding reactively to user queries, the assistant can autonomously suggest maintenance actions, highlight opportunities for performance optimization, and flag emerging operational risks. This shift from passive data retrieval to active insight generation marks a substantial improvement in decision support.

Additionally, the integration of vision capabilities enables the AI model to analyze visual content—such as equipment images, diagrams, or inspection photos—and extract relevant insights. This multimodal approach allows the assistant to combine textual and visual data, enhancing its diagnostic precision and broadening its applicability across diverse RM&D scenarios.

Even when operating within a unimodal framework, the system demonstrates the ability to link and synthesize information from multiple data sources. By aggregating telemetry, documentation, and service records into a unified conversational interface, the assistant facilitates a more coherent and efficient analysis process, reducing fragmentation and improving situational awareness.

One of the primary challenges identified in the paper relates to ensuring robust security and compliance as the AI assistant begins to interact with sensitive operational data. Addressing these concerns is essential for enabling safe, trustworthy, and scalable deployment in industrial environments. Key considerations include data privacy, auditability, user authentication, and adherence to ethical AI practices. These elements are critical not only for protecting sensitive information but also for maintaining operational integrity and regulatory compliance.

Another equally important topic is the need for continuous maintenance of the AI assistant. As the system evolves and becomes more deeply integrated into RM&D workflows, maintaining its performance and relevance requires strategic attention. This includes:

- **Model Retraining:** Regular updates to the underlying models are necessary to incorporate new data, adapt to changing operational conditions, and improve accuracy over time.
- **Knowledge Base Expansion:** Continuously enriching the assistant's knowledge base with updated documentation, service bulletins, and user feedback ensures that it remains a reliable source of insight.
- **Performance Metrics Monitoring:** Establishing and tracking key performance indicators—such as response accuracy, user satisfaction, and system uptime—helps guide iterative improvements and ensures the assistant continues to deliver value.

This shift from traditional, reactive maintenance to a strategic, data-driven approach reflects the evolving role of AI systems in industrial environments. Continuous maintenance is no longer a background task but a core component of sustaining intelligent, adaptive, and trustworthy digital services.

Proposed Solution

The foundation of AI agent for Digital Service advisory and the strategy adopted is presented in the below section.

Problem Statement: Section 2.1 present BH starting point and customer needs.

Agents landscape: Section 2.2 present a concise definition and history of AI agents.

Constraints and limitations: 2.3 present the specific risk and constraints related to the Oil&Gas industry service that motivates our approach.

Architecture: Building on this foundation Section 2.4 presents approach and methodology used for developing the Agent and explains the solution to guarantee data security, segregation and compliance.

Problem Statement

Rotating equipment, such as turbomachinery, represents some of the most critical assets in industrial operations across the Oil & Gas and Energy sectors. These machines support essential applications including Liquefied Natural Gas (LNG) processing, pipeline transport, refining, petrochemical production, and power generation. For operators, ensuring continuous availability and minimizing downtime throughout the equipment lifecycle is paramount. Unscheduled shutdowns of turbomachines can lead to full plant outages, resulting in substantial production losses [12] [13][14].

To mitigate these risks, we have developed Remote Monitoring & Diagnostic (RM&D) services, applied across a broad installed base of rotating equipment. Through its iCenter™ ecosystem, we continuously acquire sensor data from deployed assets at customer sites. This data is analyzed in near real-time using anomaly detection models, enabling early identification of emerging issues and supporting proactive maintenance strategies.

Despite these advancements, users often face significant challenges in navigating RM&D platforms. The complexity of the interfaces—characterized by multiple tabs, fragmented data views, and non-intuitive workflows—can lead to frustration, confusion, and considerable time loss. These inefficiencies hinder timely decision-making and reduce the overall effectiveness of monitoring operations. The volume and complexity of operational data grow, traditional Customer Dashboard approaches face limitations in scalability, responsiveness, and interpretability.

Agentic AI systems offer a promising solution to these challenges by embedding autonomous, goal-directed intelligence within digital service platforms. These systems go beyond traditional monitoring tools by interpreting and synthesizing data from multiple sources, delivering contextualized insights, and offering proactive recommendations through natural language interaction. Their ability to facilitate fast and accurate diagnostics, reduce manual analysis, and minimize equipment downtime positions them as strategic enablers in the pursuit of operational excellence and cost-effective maintenance.

Agents landscape

Krishnan [2] Define an AI agent as *«A system or program capable of autonomously performing tasks on behalf of a user or another system by designing its workflow and utilizing available tools».* Autonomous agents were present already before the diffusion of Large Language Models and generative AI and were usually based on simple condition-action rules (if-then statements). Today the widespread availability and adoption of Large Language Models like GPT4 have made possible a change of paradigm.

LLM-backed architecture demonstrates significant potential given the impressive performance of LLMs across language understanding and interactive decision-making tasks, as well as their capability for generating structured action sequences (e.g., through chain-of-thought prompting). LLMs have also been proved effective in programmatic interaction tasks, like generating structured function calls, *e.g.,* compliant with OpenAPI specifications, composing SQL queries for data retrieval, or orchestrating service-level operations through code-based interfaces such as Python APIs. This capability positions LLMs as versatile engines capable of bridging natural language inputs with executable actions in software environment. While conventional if-else AI systems typically operated within predefined boundaries and required explicit instruction for each task, LLM-backed AI agents demonstrate great autonomy and goal directed behavior [2] [3].

The constituting elements of an AI agent are: an LLM backend that, following [4], we will call the *brain* of the agent and that processes language inputs and generates thoughts, set of actions or function calls; a set of *tools* that connect the agent with external systems (*i.e.,* in the context of our project, an API that produces company RMD data) and an *orchestrator,* that act as a communication layer between LLM and the API's. The orchestrator can also implement some additional logic to deterministically check and integrate LLM's output. As we are going to show in 2.3.3 this has been leveraged by us to mitigate the LLM's intrinsic risk of hallucination [5] and their nondeterministic nature to guarantee data security and reliability.

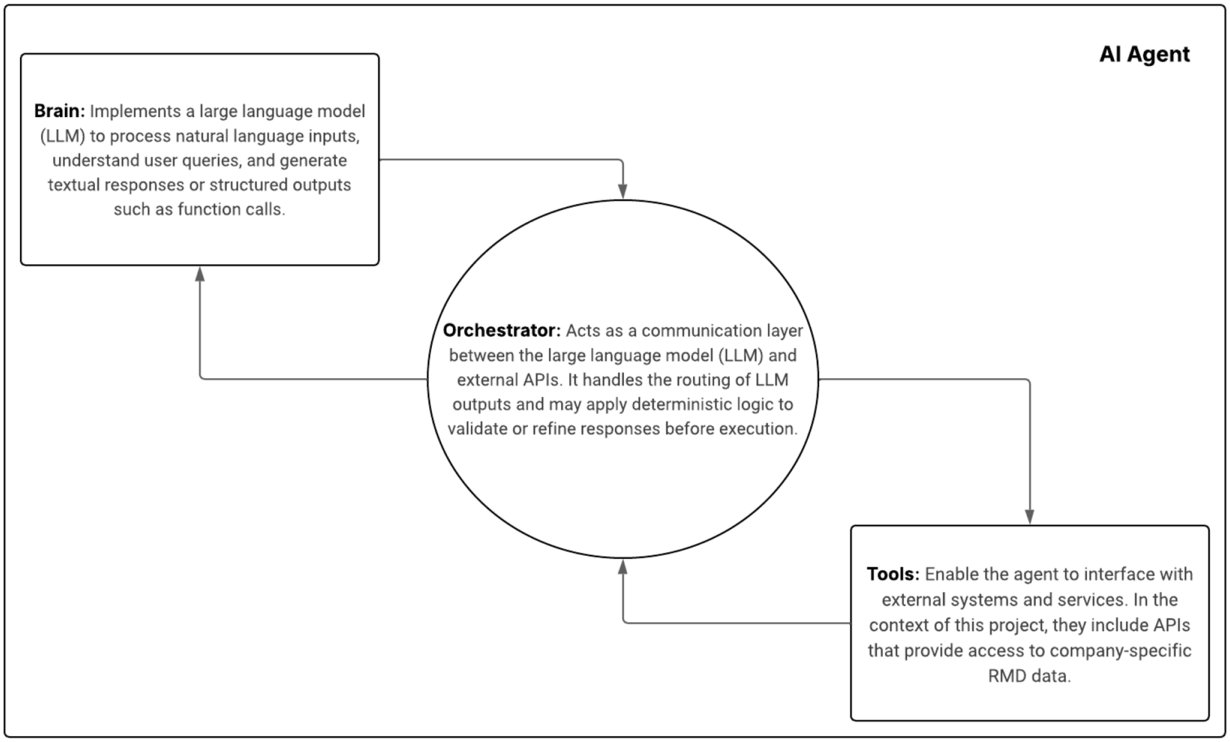

Figure 1—A conceptual overview of the core components that constitute an AI agent powered by a large language model, highlighting their roles in language understanding, decision-making, and interaction with external systems.

Constraints and limitations

AI systems present significant security challenges and pose substantial risks concerning intellectual property (IP) protection and privacy preservation. Data processed within RM&D services are classified as highly confidential, potentially containing information protected under intellectual property rights belonging to either clients or the service provider. Ensuring strict data confidentiality is essential not only to safeguard client privacy, but also to prevent the unauthorized disclosure of proprietary information. Protecting confidentiality of data and avoiding any kind of IP leakage is critical to preserving the strategic interests and competitive advantage of both the organization and its clients.

The primary intellectual property risks associated with AI systems can be categorized into two distinct threats categories: leakage-out, which refers to the unintended exposure of confidential or proprietary information and can occur either through hallucinations, which are unintentional and model-internal errors, or through adversarial attacks, which are deliberately crafted by malicious users to manipulate the system into disclosing sensitive data; and *leakage-in*, i.e. the introduction of IP protected information in the model output due to the accidental presence of such information in the model's training dataset [4].

These considerations underscore the importance of designing AI systems with robust safeguards against both leakage-out and leakage-in risks, particularly in contexts involving sensitive intellectual property and personal data. In Section 2.3, we describe how the architecture adopted for the RM&D service has been carefully crafted to address these challenges and to ensure strong data segregation.

Architecture

Several architectural frameworks and approaches have been proposed for defining LLM-based agent architectures. Anthropic [6] introduced a very influential taxonomy for LLM based agents' architectures distinguishing between two primary approaches: *Workflows* and *Agents*. *Workflows* are system where LLM and tools are orchestrated through predefined code paths (so in a fashion more similar to that of old if-else agents we mentioned in 2.1). In workflow architectures, tasks are decomposed into sequential steps, with each LLM call processing the output of the previous step following a deterministic execution path. This rule-based approach minimizes the risk that the LLM component could deviate from intended behavior due to hallucinations and return erroneous responses to users [16]. However, for this approach to be feasible, the operational scenarios must be easily decomposable into fixed subtasks. This is not applicable in our RM&D use case as our system must be able to navigate different data sources depending on user inputs and must be able to dynamically change its path and search for new data if the current situation requires it to do so.

Consequently, we selected the second family of architectures proposed by Anthropic - *Agents:* systems where LLMs dynamically orchestrate their own process and tool usage. Agents define their task through interaction with the user and, once the task is clear, plan and operate independently getting their "ground truth" from the environment through *tool* usage or potentially returning to the user for further information or clarification. This makes agents perfectly suitable for open-ended tasks like those we need to address in our RM&D service context. At the same to guarantee satisfying performances and prevent hallucinations the system and the toolset and their documentation must be clearly and thoughtfully defined and some.

Recent work [7] builds a further distinction among *AI agents,* designed as single entity systems (*i.e.* with a single LLM *brain)* and *Agentic AI* systems, *i.e.* systems composed of multiple specialized agents that coordinate, communicate and collaborate to achieve a common goal.

Agentic AI, despite demonstrating considerable potentials, still poses major risk and limitations mainly due to *inter-agent distributional shift* (*i.e.* when the behavior of one agent alters the operational context for others) and *compound error* (*i.e.* when a faulty or hallucinated output from one model propagate through the system corrupting subsequent decision). Having an ensemble of Agents also enhances the risk intrinsic to their black box nature and expand the attack surface from a security and adversarial risk point of view compared to a single agent architecture [4]. Our project required careful balance between innovation to accommodate diverse use cases and mitigation of inherent technological risks, given the sensitivity of processed data and the critical economic and security value of monitored assets: therefore, we decided to opt for a single *AI Agent* architecture.

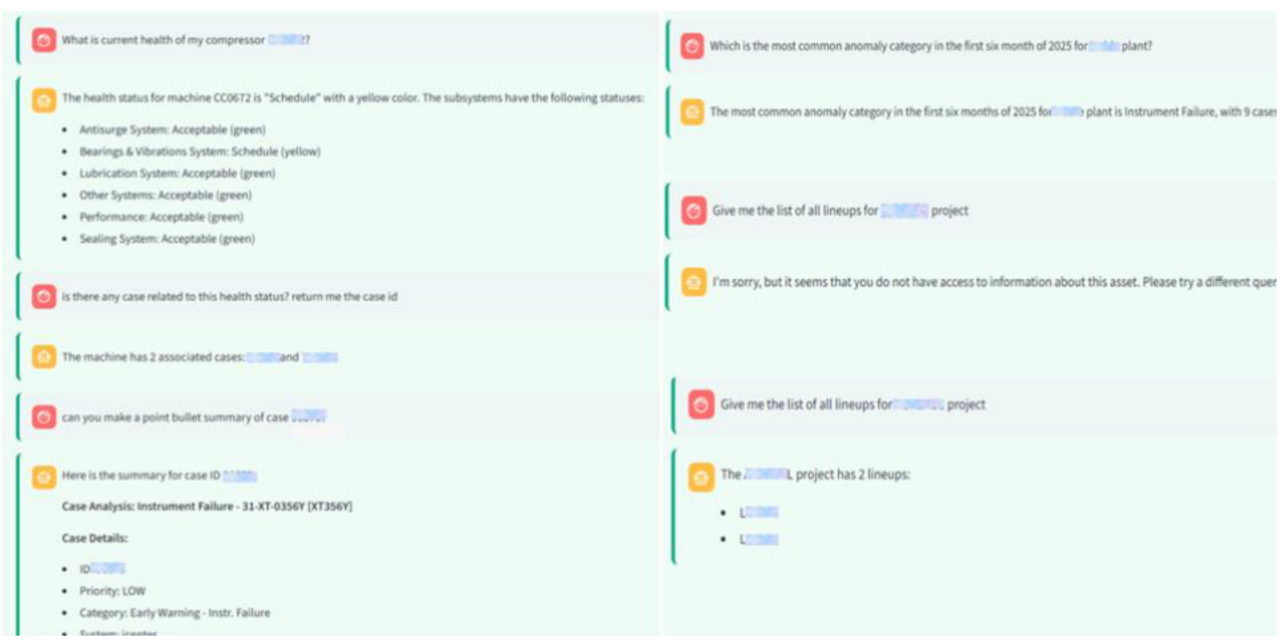

Figure 2—Illustrative example of an interaction with the Agent demonstrating its ability to retrieve and process relevant information from the RM&D database, such as summarizing open cases or aggregating information from different sources. The figure also highlights the system's data segregation capabilities: the same query regarding a specific plant was submitted by two users with different access privileges. The agent correctly returned information only to the authorized user, while appropriately restricting access for the unauthorized one.

Brain

As outlined in Section 2.2, our system requires complete isolation to eliminate leakage-out risks. Using closed source LLMs or any kind of service that allows us to interact with an LLM engine through an external API has to be excluded as this implied sending potentially IP-protected High confidentiality data to external entities. We therefore implemented an in-house solution utilizing an open-source LLM deployed on corporate premises. The selected model is an instance of Mistral Small 24b-Instruct: this model can achieve state-of-the-art performances on many benchmarks and, in particular, presents great capabilities in agentic tasks like function calling, JSON outputting and reasoning [8]. At the same time this model falls in the category of "small" Large Language Models below 70b of parameters, enabling local deployment and inference without requiring extensive computational resources. In our case the model was deployed on a corporate-owned NVIDIA DGX workstation.

The model implementation follows the ReAct framework [9], which prompts LLMs to generate both verbal reasoning traces and actions pertaining to a task in an interleaved manner. The ReAct framework implements a loop where each cycle consist of a sequence of: Thought Generation (generation of reasoning traces based on current context) → Action Selection (tool calling based on previous reasoning) → Observation (Incorporation of Tools response into model context or agent memory). This paradigm enables LLM agents to tackle complex, multi-step tasks requiring both reasoning and interaction with external systems making them suitable for autonomous problem-solving scenarios.

The prompt design explicitly incorporates constraints aimed at mitigating leakage-in risks. It instructs the model to generate responses exclusively based on information retrieved through tool outputs, thereby avoiding reliance on its pre-trained internal knowledge representations.

Tools

The RM&D service processes diverse data types, including time series, alarms, events, structured tabular data, and unstructured textual information, such as email exchanges with clients or event/anomalies analysis. These data are managed through the Digital Service Data Layer, an interface comprising APIs and connectors that decouple data sources from applications and services requiring data access. Anyway, while the current Data Layer provides a functional abstraction for interfacing with underlying databases, its design is primarily oriented towards data connectivity rather than intelligent orchestration and like most current enterprise API, as outlined in [10], is designed for human-driven interaction patterns or machine to machine interaction but not of the goal-oriented dynamic kind provided by AI agents.

For this reason, specific endpoints have been developed to support agentic workflow. The endpoints were developed with an *intent-based* design [10] allowing agent to perform complex tasks through highly abstract operation rather than multiple granular calls. This allows for easier interaction of the model with the API both in terms of the number and kind of requests that agent must generate to get the necessary data and in terms of number of requests that the backend has to process. Additionally, having dedicated API endpoints for agentic interaction enables the implementation of custom API policies for security and segregation (as explained in Section 2.3) and supports enhanced documentation, error handling, and metadata management to enrich agent context and facilitate correct endpoint usage and response interpretation.

Although our current architecture is structured around a single-agent interaction model, this does not preclude the implementation of tools that leverage generative AI capabilities not as autonomous decision-making engines, but rather as specialized NLP components. Given that RM&D data encompasses unstructured information (e.g., potentially extensive email exchanges between clients and service providers regarding specific issues), a dedicated endpoint was developed that leverages the same LLM engine used by the Agent (but of course tailored to the specific summarization task by prompt engineering) to provide a summarization endpoint capable of returning summaries for unstructured textual RM&D data.

Orchestrator

The orchestrator serves as the central coordination and control layer within an LLM agent architecture, it acts as an intermediary system that manages the interaction flow and execution sequencing between LLM brain and the tools available in its execution context. It implements the logic required to route API calls generated by the LLM, manage execution workflows, and enforces system constraints. In our implementation, the orchestrator has been implemented using Python and the LangGraph [11] framework. The orchestrator is also responsible for managing memory operations, including the coordination of data storage, retrieval, and contextual integration across interactions. This functionality ensures consistent state tracking and supports the agent's ability to maintain coherence over multi-turn tasks.

As a deterministic module, the orchestrator layer can be leveraged as a deterministic bridging mechanism that mitigates the inherent nondeterministic behavior of LLM outputs. As explained in 2.2 our context required us to eliminate the risk of data leakage and it was necessary to enforce a strict data segregation in order to prevent scenarios where (*e.g.* because of a hallucination of the LLM) client *X* could receive data belonging to client *Y.* A dedicated layer within the corporate API infrastructure is already in place to enforce data segregation and access control policies. However, this mechanism relies primarily on the credentials of the end user to determine authorization levels, we needed an effective way to manage this access logic.

Security

Our solution builds upon the existing API authorization layer, which provides a robust foundation for enforcing access policies at the infrastructure level and leverage on the orchestration layer to completely decouple authorization and segregation logic and generative AI components. Sensitive information—such as user identifiers or asset visibility scopes—is never exposed to, nor processed by, any generative module. This strict architectural decoupling ensures that access control and data protection are enforced externally and reliably, thereby eliminating the risk of leakage-out due to model hallucinations or adversarial prompt injections.

This architectural approach prevents malicious users from executing *prompt injection attacks* against our agent (*i.e.* manipulating model prompt to guide the language model toward an unwanted behavior, such as bypassing data segregation), as the LLM has no role in managing user access; Similarly a malicious user cannot steal user credential by a *prompt leaking attack,* as no authentication information is ever present in the model context. See [4] for a detailed survey of LLM agent's threats.

This architectural strategy not only reinforces the system's resilience against common LLM-specific vulnerabilities, but also exemplifies a principled separation of concerns—ensuring that access control and data segregation remain fully deterministic and verifiable, while generative components are confined strictly to their intended scope of operation

As the assistant begins to interact with sensitive operational data, ensuring robust security and compliance becomes critical:

- Data Privacy: Implement strict access controls and encryption protocols to protect user and equipment data. Ensure compliance with GDPR, ISO/IEC 27001, and other relevant standards.
- Auditability and Explainability: Ensure that the assistant's decisions and recommendations are traceable and explainable. This is especially important in regulated industries where accountability is key.
- User Authentication and Role-Based Access: Introduce secure login mechanisms and role-based permissions to ensure that users only access data and functionalities appropriate to their responsibilities.
- Ethical AI Practices: Establish guidelines for responsible AI use, including bias mitigation, fairness, and transparency in decision-making.

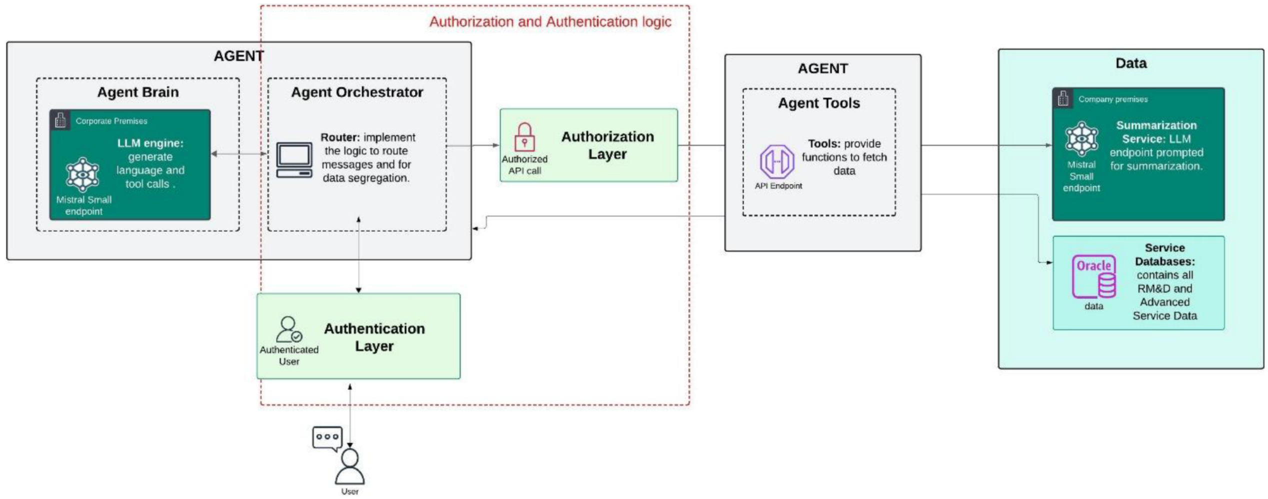

Figure 3—Agent components are shown in grey, while dark green elements represent LLM endpoints. Notably, the authentication and authorization logic are fully decoupled from the LLM services, ensuring safety and separation of concerns.

Result

Validation of the proposed architecture was conducted using a manually annotated dataset comprising 70 use cases, each accompanied by a verified ground truth response. These test cases consisted of carefully constructed prompts designed to evaluate both the accuracy of the model's responses and its susceptibility to information leakage. Each output was systematically assessed across three key dimensions: *correctness, leakage-out,* and *leakage-in.*

The use cases were defined in collaboration to final user to test the responses on the major topics and interaction the Agentic AI could establish in the specific context. Using the test set for understating the possibility to link the output of RM&D and interconnection between all the additional service available. The test output guarantees the validity of the paper intention and the possibility to define future next step.

Correctness was evaluated in binary terms, given the presence of objective ground truth and the specificity of the queries. A response was deemed *correct* if it aligned with the user's intent and matched the ground truth in both informational content and tone; conversely, it was classified as *incorrect* if it contained factual inaccuracies or hallucinated content. To assess leakage risks, the evaluation included targeted adversarial scenarios. Specific prompts were crafted to simulate potential jailbreak attempts, such as instructing the model to impersonate another user or disclose credential-related information. These tests were designed to probe the model's robustness against *adversarial attacks.*

The agent exhibited satisfactory performance in generating accurate responses across the evaluated prompt set. Notably, the architecture demonstrated strong resilience against information leakage vulnerabilities, with no instances of either leakage-in or leakage-out observed across the entire test set. In terms of accuracy, the agent produced correct responses in 82.86% of the evaluated cases. This result is already indicative of a robust and reliable system; however, it also highlights specific areas where further refinement and enhancement of the agent's capabilities could lead to improved performance and broader applicability.

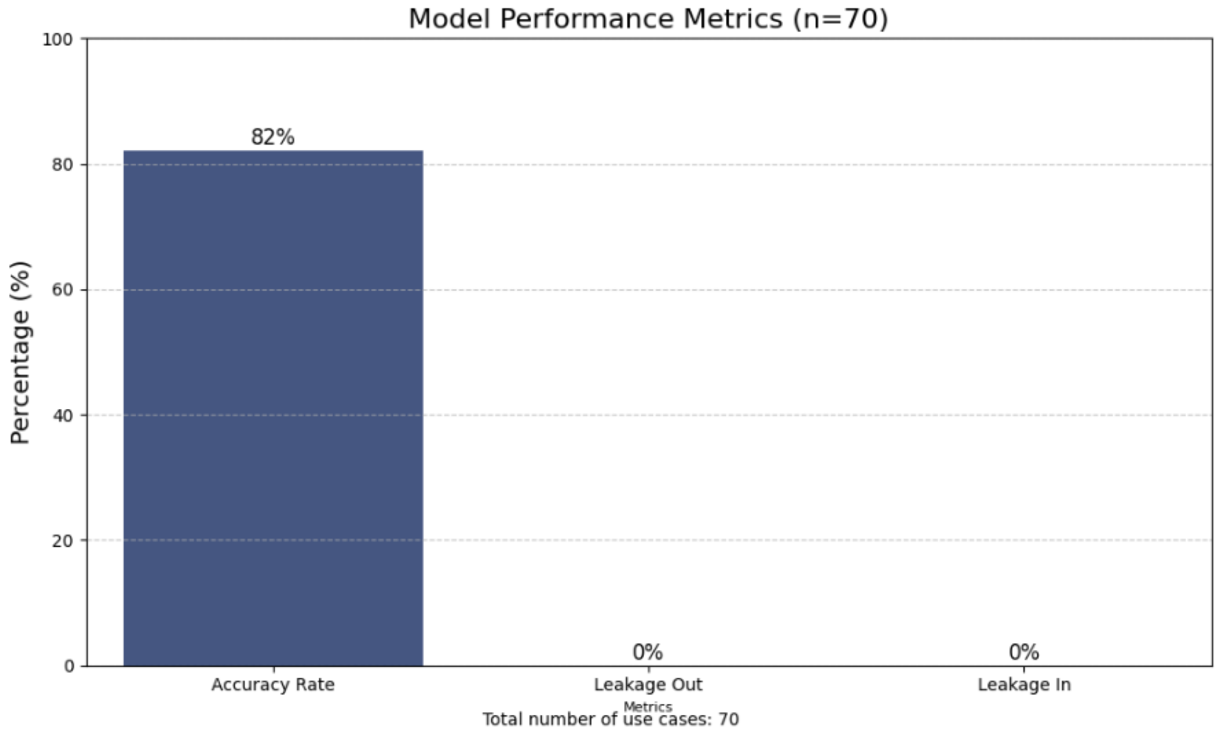

Figure 4—As part of the preliminary evaluation, the model was tested on a curated dataset comprising 70 use cases. The agent demonstrated a response accuracy of 82%, successfully generating correct answers aligned with the ground truth. Notably, throughout the evaluation, the model did not exhibit any instances of unauthorized data extraction from its internal knowledge representation, nor did it disclose sensitive or restricted information, indicating strong adherence to confidentiality and data protection standards

To complement the automated assessment of the agent's performance, a structured *human evaluation* will be conducted in future stages of the study. For this purpose, a dedicated data rubric has been developed to guide evaluators in systematically reviewing model outputs. Each response will be assessed across four critical dimensions: *Accuracy,* rated on a 3-point scale to capture factual correctness and precision; *Faithfulness,* evaluated on a 5-point scale to measure the consistency and relevance of the response with respect to the input and source material; and the two binary dimensions of: *leakage-in* and *leakage-out* to assess the presence of leaked IP protected or unauthorized information in the response.

The integration of an Agentic AI system capable of linking and synthesizing data from multiple sources offers significant benefits to customers in RM&D applications. By unifying telemetry, maintenance records, service bulletins, and visual inspections into a single, coherent interface, the assistant eliminates the need for manual data gathering across fragmented platforms. This not only streamlines workflows but also drastically reduces time spent searching for relevant information—time that can instead be invested in strategic decision-making.

One of the most impactful advantages is the acceleration of problem resolution. By providing immediate access to contextualized insights and historical data, the assistant enables faster root cause identification and more accurate diagnostics. This leads to quicker maintenance interventions and a substantial reduction in equipment downtime. For engine systems, where every hour of inactivity can translate into significant operational and financial losses, the ability to act swiftly is critical.

Moreover, the assistant's proactive recommendations and natural language interface make complex data accessible to a broader range of users, reducing reliance on specialized analysts and empowering frontline personnel. The result is a more agile and responsive maintenance ecosystem, where issues are addressed before they escalate, and operational continuity is preserved.

To ensure continued relevance and performance improvement, the Agentic AI system must be equipped with mechanisms for continuous learning, where user feedback plays a central and transformative role.

By actively engaging users in the refinement process, the system evolves from a static tool into a dynamic, adaptive assistant. Key components of this learning framework include Feedback Loops; Adaptive Learning; Model Retraining Pipelines; Knowledge Base Expansion; Monitoring and Evaluation.

In this framework, user feedback is not merely a supplementary input but a foundational element that drives the assistant's evolution, enabling it to deliver increasingly precise, contextual, and valuable support in RM&D environments.

Conclusions

This research demonstrates how Agentic AI can transform Remote Monitoring & Diagnostics (RM&D) services from passive data visualization tools into intelligent, conversational partners for industrial operators. The presented work addresses a critical gap in turbomachinery monitoring by enabling natural language interaction with complex operational data while maintaining strict security requirements.

Single-agent architecture successfully balances operational flexibility with enterprise security constraints. Deployment of open-source LLMs on corporate premises, combined with security architecture, shows that organizations can use advanced AI without exposing intellectual property to external services.

The paper demonstrates that advanced AI capabilities can be securely integrated into sensitive industrial applications without compromising data integrity. The architectural principles—particularly separating authorization from AI components—apply beyond industrial monitoring to any domain requiring AI systems to operate with sensitive data under strict access controls. While our single-agent approach avoids the complexity and security risks of multi-agent systems, future work could explore controlled agent collaboration for specialized analytical tasks.

The shift from reading dashboards to conversing with data significantly reduces cognitive load on operators and accelerates decision-making. In turbomachinery applications, where rapid response prevents failures and production losses, this improvement has direct economic value. By transforming industrial monitoring from reactive analysis to proactive intelligent partnership, this research establishes a foundation for safer, more efficient industrial operations while providing practical guidance for secure AI implementation in critical infrastructure.

The beneficial effects for customers enabled by Agentic AI systems are multifaceted, ranging from significant time savings to enhanced decision-making support. By intelligently linking and synthesizing data from multiple sources—such as telemetry, maintenance logs, service documentation, and visual inspections—the assistant eliminates the need for manual data retrieval across fragmented platforms. This streamlining of information access reduces time wasted in navigating complex dashboards or searching for relevant insights.

Moreover, the assistant's ability to deliver contextualized, proactive recommendations empowers users to make faster and more informed decisions. In operational environments where equipment uptime is critical—such as in engine systems—this capability translates directly into reduced downtime. By accelerating fault detection, root cause analysis, and maintenance planning, the assistant helps prevent prolonged outages and minimizes the operational and financial impact of failures.

In essence, the Agentic AI system transforms RM&D dashboards from passive data viewers into active, intelligent collaborators. Customers benefit not only from improved efficiency and responsiveness but also from a more strategic approach to asset management, where data becomes a driver of performance, reliability, and long-term value.

The single-agent approach also lays the foundation for future scalability, enabling the deployment of intelligent assistants across fleets, facilities, and service domains with minimal reconfiguration. It supports continuous learning, cross-domain generalization, and strategic decision-making—positioning the Agentic AI system not merely as a tool, but as a foundational component of next-generation digital services.

Future enhancements will consider learning from user feedback in order to improve the efficiency and the robustness of the agent in a production environment.

References

1. A. H. et al, «"Rotating machinery prognostics: State of the art, challenges and opportunities," "Rotating machinery prognostics: State of the art, challenges and opportunities,",» *Mechanical Systems and Signal Processing*, vol. **23**, n. 3, pp. 724–739, 2009.
2. N. Krishnan, «AI Agents: Evolution, Architecture, and Real-World Applications,» arXiv preprint arXiv:2503.12687, 2025.
3. AWS, «What Are AI Agents? Amazon Web Services,» 2024. [Online]. Available: https://aws.amazon.com/what-is/ai-agents/.
4. Z. D. et al, «AI Agents Under Threat: A Survey of Key Security Challenges,» arXiv:2406.02630v2, 2024.
5. Y. B. et al, «HalluLens: LLM Hallucination Benchmark,» arXiv:2504.17550, 2025.
6. Anthropic, «Building effective agents,» 2024. [Online]. Available: retrieved from: https://www.anthropic.com/engineering/building-effective-agents.
7. R. S. et al, «AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges,» arXiv:2505.10468, 2025.
8. Mistral, «Mistral-Small-24B-Instruct-2501,» 2025. [Online]. Available: https://huggingface.co/mistralai/Mistral-Small-24B-Instruct-2501.
9. S. Y. et al, «REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS,» arXiv:2210.03629v3, 2023.
10. S. T. Vaibhav Tupe, «AI Agentic workflows and Enterprise APIs: Adapting API architectures for the age of AI agents,» arXiv:2502.17443, 2025.
11. LangGraph, «https://www.langchain.com/langgraph,» 2025. [Online].
12. Iannitelli, M., Allegorico, C., Garau, F., & Capanni, M. (2018). A hybrid model for on-line detection of gas turbine lean blowout events. In Phm society european conference
13. Ghiasi S., Schillaci, G., Ricci, A., Allegorico, C. and Veneri, Giacomo, Temporal Graph and Generative Modeling for Unsupervised Anomaly Detection of Industrial Sensor Systems: A Real World Application in Turbo-Machines. Available at SSRN: https://ssrn.com/abstract=4983865 or http://dx.doi.org/10.2139/ssrn.4983865
14. V. Gori, G. Veneri and V. Ballarini, "Continual Learning for anomaly detection on turbomachinery prototypes - A real application," 2022 IEEE Congress on Evolutionary Computation (CEC), Padua, Italy, 2022, pp. 1–7, doi: 10.1109/CEC55065.2022.9870234.
15. D. B. Acharya, K. Kuppan and B. Divya, "Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey," in *IEEE Access*, vol. **13**, pp. 18912–18936, 2025, doi: 10.1109/ACCESS.2025.3532853.
16. Gaurang Sriramanan, Siddhant Bharti, Vinu Sankar Sadasivan, Shoumik Saha, Priyatham Kattakinda, Soheil Feizi, LLM-Check: Investigating Detection of Hallucinations in Large Language Models, Part of Advances in Neural Information Processing Systems **37** (NeurIPS 2024) Main Conference Track